PRACTICAL 3

Practical3(1)

Problem Phase

This problem is about implementing and comparing three sorting algorithms: Bubble Sort, Selection Sort, and Insertion Sort. The aim is to arrange student marks in ascending order.

Solution

Bubble Sort compares adjacent elements and swaps them if they are in the wrong order. Selection Sort finds the smallest element and places it in the correct position. Insertion Sort takes each element and inserts it into its correct position in the sorted part of the array. All three methods produce the sorted marks.

Practical3(2)

Problem Phase

This problem is about **sorting an array containing only 0, 1, and 2** in ascending order without using extra storage.

Solution

The program uses three pointers: low, mid, and high. low is used for 0s, mid checks the current element, and high is used for 2s. Using swaps, all 0s are placed first, 1s in the middle, and 2s at the end in **one pass** through the array.